Jackson Rader

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EDUCATION

Boston University, College of Engineering Boston, MA
Bachelor of Science in Mechanical Engineering ***Expected May 2028***
GPA: 3.30 / 4.00
Relevant Coursework: Engineering Design, Energy & Thermodynamics, Mechanics, Circuits

ENGINEERING EXPERIENCE

Boston University Terrier Ride Engineering Club Boston, MA
Team Build Member *September 2026 – Present*
- Design and fabricate track and structural components for a scale-model roller coaster entered in a collegiate engineering competition
- Take parts from 3D CAD models to finished components through machining and manufacturing processes

Boston University Rocket Propulsion Group Boston, MA
Member *September 2025 – Present*
- Machine metal components from mechanical designs and simulations for a team aiming to launch the first collegiate liquid-fueled rocket to space
- Apply fluid properties and propulsion fundamentals to liquid rocket engine design and propulsion system components

Pickard Chilton New Haven, CT
Architecture Apprentice *September 2023 – May 2024*
- Built physical and digital models to communicate design concepts for commercial projects, following them from concept through final construction

PROJECTS

Animated Desktop Robot | *Fusion 360, Arduino, 3D Printing* *In Progress*
- Designing a desk-sized robot end to end, owning the mechanical CAD, 3D-printed shell, wiring, soldering, and embedded code
- Integrating an 8x8 LED matrix for facial expressions, 3 servos for head and arm motion, a text-to-speech audio module, and rechargeable power

Adaptable Whack-A-Mole Device | *Team Project*
- Co-designed an assistive whack-a-mole device with adjustable difficulty to support motor skill development in children

Point-to-Point Wireless Communication Device | *Team Project*
- Co-designed and prototyped a pair of wireless devices that sent and displayed messages at a five-foot range

Raspberry Pi Event Photo Booth | *Python, Raspberry Pi*
- Built a Python photo booth application for live events with camera auto-detection, custom themed templates, and QR code photo delivery

LEADERSHIP

Tau Kappa Epsilon, Boston University Boston, MA
Former President; previously Treasurer and Historian
- Led the chapter as President after serving as Treasurer, managing chapter finances, and as Historian
- Raised over $10,000 for St. Jude Children's Research Hospital by organizing chapter charity events
- Designed social media promotion for chapter events, including sponsored content with Red Bull, Spikeball, and Raising Cane's

SKILLS

CAD & Design: SolidWorks, Onshape, Fusion 360, AutoCAD, Revit
Prototyping & Fabrication: Milling, 3D Printing, Laser Cutting, Soldering, Circuit Prototyping
Programming & Electronics: MATLAB, Python, Arduino, Raspberry Pi, Oscilloscope